Jintao Li 李锦韬

PhD Researcher · Intelligent Optimization and Decision-Making

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I study intelligent optimization for complex decision problems. My work draws on reinforcement learning and evolutionary computation, with an emphasis on multi-objective and constrained optimization. I apply these methods to analog electronic design automation and large-model post-training.

RESEARCH FOCUS

Intelligent Optimization

Learning-based search for expensive problems with competing objectives and hard constraints, using reinforcement learning and evolutionary computation.

Analog EDA

Circuit sizing, behavior-centric representation, PVT-robust design, and industrial EDA integration.

Large-Model Post-Training

Comparison feedback, Pareto-front improvement, preference optimization, and constrained policy learning.

EDUCATION & RESEARCH EXPERIENCE

2024.09–2027.06 **PhD Researcher**, Shenzhen Institute for Advanced Study, UESTC. Computer Science and Technology. Expected completion: June 2027.

2021.09–Present **Prof. Yun Li's Research Group**, UESTC. Research on intelligent optimization, especially reinforcement learning and evolutionary methods for multi-objective problems, with applications in analog EDA and large-model post-training.

2021.09–2024.06 **MEng, Electronic Information**, Shenzhen Institute for Advanced Study, UESTC.

2016.09–2020.06 **BEng, Optoelectronic Information Science and Engineering**, Guangzhou University.

SELECTED PUBLICATIONS (1–6)

1. **Learning from Comparison: Constrained Projection Policy Optimization for Pareto-Front Improvement.**
   Jintao Li, Maowen Tang, Yongji Long, Weixuan Liu, Yanlang Zheng, Sicheng He, Ao-Jin Li, Shui Yu, Yun Li. **ICML 2026 · CCF-A.**
2. **Decoupling Analog Circuit Representation from Technology for Behavior-Centric Optimization.**
   Jintao Li, Haochang Zhi, Jiang Xiao, Keren Zhu, Yun Li. **DAC 2025 · CCF-A / EDA flagship.**
3. **AnalogGym-Opt: An LLM-Oriented Optimization Infrastructure for Analog Circuit Sizing with GRPO.**
   Jintao Li, Haochang Zhi, Yongji Long, Weixuan Liu, Yanhan Zeng, Keren Zhu, Shui Yu, Yun Li. **IEEE TCAD, 2026 · CCF-A.**
4. **A Behavior-Centric Methodology for High-Performance Amplifier Sizing.**
   Haochang Zhi†, **Jintao Li**†, Keren Zhu, Yun Li, Weiwei Shan, Wanyuan Qu. **IEEE TCAD, 2026 · CCF-A.**
5. **Archive-Protected Survival Selection for Response Evolution in Large Language Model Post-Training.**
   Jintao Li, Yongji Long, Weixuan Liu, Aojin Li, Yanhan Zeng, Shui Yu, Yun Li. **IEEE TEVC, 2026 · CCF-B.**
6. **Priority-Based Graph-Enhanced Reinforcement Learning for Robust Analog Circuit Optimization.**
   Jintao Li, Zhenxin Chen, Sicheng He, Ao-Jin Li, Shui Yu. **AAAI 2026 · CCF-A.**

SELECTED PUBLICATIONS (7–11)

7. **Knowledge Transfer Framework for PVT Robustness in Analog Integrated Circuits.**
   Jintao Li, Yanhan Zeng, Haochang Zhi, Jingci Yang, Weiwei Shan, Yongfu Li, Yun Li. **IEEE TCAS-I 71(5), 2017–2030, 2024.**
8. **Multitask Evolution with Problem Reformulation for Global Exploration in Analog Circuit Design.**
   Jintao Li, Sicheng He, Aojin Li, Shui Yu, Yun Li. **Advanced Engineering Informatics 70, 104195, 2026.**
9. **AnalogGym: An Open and Practical Testing Suite for Analog Circuit Synthesis.**
   Jintao Li, Haochang Zhi, Ruiyu Lyu, Wangzhen Li, Zhaori Bi, Keren Zhu, Yanhan Zeng, Weiwei Shan, Changhao Yan, Fan Yang, Yun Li, Xuan Zeng. **ICCAD 2024 · EDA flagship.**
10. **Multi-Task Evolutionary to PVT Knowledge Transfer for Analog Integrated Circuit Optimization.**
    Jintao Li, Haochang Zhi, Weiwei Shan, Yongfu Li, Yanhan Zeng, Yun Li. **ICCAD 2023 · EDA flagship.**
11. **Balancing Objective Optimization and Constraint Satisfaction for Robust Analog Circuit Optimization.**
    Jintao Li, Haochang Zhi, Jiang Xiao, Yanhan Zeng, Weiwei Shan, Yun Li. **ASP-DAC 2025, 190–196.**

† Equal contribution (co-first authors).

RESEARCH SYSTEMS

Analog-IC EDA Software Stack — Lead Developer
A mature workflow connecting a server-side optimizer, an approximately 10-GB circuit dataset, a lightweight GUI, and industrial simulation tools.

3 circuit classes · 4 tapeout validations · 22 topologies · 4 technology nodes

Self-Evolving Analog Circuit Design Agent — Ongoing Team Project
An ongoing framework for dynamic agent routing, comparison-based GRPO, constrained policy optimization, trajectory replay, reusable skills, simulation evaluation, and human review.

Dynamic agent routing · Comparison feedback · LLM post-training · Design memory

AnalogGym — Open Benchmark
An open testing suite for reproducible evaluation of analog synthesis methods across five circuit categories, multiple process technologies, and both commercial and open-source simulators.

Paper: arXiv 2409.08534 · Code: github.com/CODA-Team/AnalogGym

HONORS & AWARDS

2026.10	**National Scholarship for Graduate Students**, UESTC.
2024–2025	**National Scholarship for Graduate Students**, UESTC.
2025	**CAST Young Elite Scientist Sponsorship Program — Doctoral Track**.
2025	**UESTC Outstanding Student Nominee**, selected among 20 university-wide candidates.
2023–2025	**First-Class Scholarship**, UESTC.
2023–2025	**Outstanding Graduate Student Enterprise Scholarships**, ZTE and BYD, UESTC.
2022	**Second-Class Scholarship**, UESTC.

RESEARCH LINKS

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